

Higher quality parent-offspring relationships predict lower-risk adolescent sexual behavior: A
genetically informed sibling comparison

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Conflict of interest disclosure

All authors declare that they have no real or perceived conflicts of interest related to the conduct or publication of this research.

Clinical trial registration

Not applicable.

List of abbreviations

A – Additive genetic effects
C – Shared environmental effects
E – Non-shared environmental effects and measurement error
PEQ – Parent Environment Questionnaire
 r_{GE} – Gene-environment correlation

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Abstract

Purpose: We used data from adopted and non-adopted siblings to separate genetic and environmental confounding of the effect of parent-offspring relationships on offspring's risky sexual behavior.

Methods: Data came from the Sibling and Interaction Behavior Study (SIBS), which included 617 two-offspring families enrolled between 1998 and 2004 (53% European, 39% Korean, 8% other), with either one or two adopted offspring ($n = 409$) or two non-adopted offspring ($n = 208$). At one assessment (M age = 17.3), parents and offspring reported on parent-offspring relationship quality; at a later assessment (M age = 20.7), offspring reported their sexual behavior. Univariate biometric models tested shared environmental effects and mixed-effects models examined between- and within-family effects.

Results: Biometric modeling indicated that 15% of the variance in sexual behavior reflected shared environmental effects. Prospectively assessed parent-offspring relationship quality was significantly associated with offspring sexual behavior in both non-adopted and adopted families, eliminating the possibility of genetic confounding, and within as well as between families, consistent with a causal effect. Results were similar across both parent and offspring ratings of relationship quality.

Discussion: We found consistent evidence for an environmental explanation: higher quality parent-offspring relationships predicted that offspring were less likely to engage in risky sexual behavior. These associations existed both between families, reflecting the impact of shared environmental factors, and within families, consistent with environmental influences that operate within families. Future research should use research designs that separate potential confounding

from reactive gene-environment correlation and identify mechanisms underlying this association to inform interventions.

Keywords: parent-offspring relationship quality, sexual development, adoption study, gene-environment correlation, environmental influence, shared environment, sibling comparison

Implications and Contributions Statement: This study demonstrates that the association between parent-offspring relationship quality and adolescent sexual behavior is consistent across informants and robust to genetic and shared environmental confounding. Results indicate that supportive, low-conflict, parent-offspring relationships may function as a meaningful environmental influence on adolescents' sexual behavior.

Higher quality parent-offspring relationships predict lower-risk adolescent sexual behavior: A genetically informed sibling comparison

Research emphasizing environmental effects consistently demonstrates that across multiple dimensions of the parent-offspring relationship—such as warmth, closeness, and conflict—are associated with sexual health outcomes including delayed sexual initiation, fewer sexual partners, less engagement in casual sex, and lower rates of unintended pregnancy and sexually transmitted infections [1–6]. This body of research provides diverse evidence using a variety of designs, measures, and informants that supportive, low-conflict parent-offspring relationships are associated with lower engagement in risky sexual behavior.

Despite the clear strengths of existing research, there are limitations that warrant further investigation. For instance, studies largely rely on statistical control of environmental variables (e.g., socioeconomic status), rather than quasi-experimental designs, such as sibling comparisons that account for confounding from shared family environments. Arguably the most important limitation of previous research is a lack of consideration for potential genetic confounds. If the association between parent-offspring relationship quality and offspring sexual behavior can be accounted for by genes, the different theoretical perspectives positing environmental effects (e.g., [7-9]) will have to reconsider the impact of the environment. As far as we know, the issue of genetic confounds has never been addressed in the literature—an issue the present study seeks to address.

When considering associations between parenting and offspring outcomes, genetic confounds are manifested via gene-environment correlations (r_{GE}) [10,11]. We focus on two (of three) types of r_{GE} —passive and reactive. In passive r_{GE} , the environment created by parents correlates with the genes they pass on to their offspring. Imagine parents who have a genetic

predisposition toward impulsiveness and poor emotional regulation. These traits, for which parents transmit a genetic propensity to their offspring, may also lead parents to create a more conflictual and less supportive parent-offspring relationship. Their offspring, who inherit the same genetic predispositions, may later engage in risky sexual behavior due to their genetic liability rather than the rearing environment, which although associated with risky sexual behavior is not necessarily a cause of that behavior. Thus, what appears to be an environmental effect of parenting may reflect shared genetic liabilities rather than true environmental causation. Passive *r*GE has been documented widely in behavioral genetic research [12], and even small amounts of *r*GE can confound phenotypic associations [13].

Reactive *r*GE occurs when offspring heritable behaviors elicit different reactions from their parents. For example, although parents may not know whether their offspring engages in risky sexual behavior, such behavior is correlated with general disinhibition [14]. Therefore, it is plausible that an offspring's tendency to engage in delinquent behavior more broadly—such as substance use, truancy, and other misbehaviors that are more observable to parents than sexual activity—may evoke more conflictual parental responses. In such cases, the direction of causality is offspring-to-parent, rather than parent-to-offspring. Although reactive *r*GE is not directly estimated in the present analyses, because parent-offspring relationship quality was measured prior to offspring sexual behavior, it is unlikely that offspring behavior evoked the parental responses captured in these measures¹.

¹ The third type of *r*GE is active *r*GE and (like reactive *r*GE) implies an offspring-to-parent direction of causality. Active *r*GE occurs when offspring actively seek out environments that align with their genetic propensities. Some have suggested that active *r*GE processes may be less likely for effects of this kind, and outside of experimental settings they are difficult to distinguish empirically from reactive *r*GE [15]. For these reasons, we focus our discussion on reactive *r*GE.

Current Study

The present study triangulates multiple lines of evidence to test the hypothesis that higher quality parent-offspring relationship quality environmentally reduces adolescents' sexual behavior [16]. First, we undertake biometric analysis of the resemblance of adopted (i.e., do *not* share segregating alleles with parents) and non-adopted (i.e., share segregating alleles with parents) siblings to determine whether there are shared environmental effects on sexual behavior. Second, we compare parent-offspring resemblance in adoptive and non-adoptive families to investigate whether the association of parenting with offspring sexual behavior is attributable to genetic confounding. Last, we decompose the associations into between- and within-family parts to determine whether sibling differences in parent-offspring relationship quality (which are not confounded by the shared environment) are associated with differences in sexual behavior [17,18]. Further, our prospective design allowed us to reduce (but not eliminate) the possibility that associations were produced by reverse causation (e.g., reactive *r*GE).

Methods

We conducted analyses in *R*. Scale reliabilities were estimated using the *psych* package [19]. Research was approved by the University of Minnesota Institutional Review Board in accordance with the ethical principles of the Belmont Report.

Participants

We used data from the Sibling Interaction and Behavior Study (SIBS), which included 617 families with two offspring. We report relevant details of the sample here; additional information about the sample can be found elsewhere [20]. The full sample at intake (Time 1) contained 285 families with two adopted offspring, 208 families with two non-adopted offspring and 124 families with one non-adopted offspring and one adoptive offspring. Adoptive families

($N = 409$) were recruited via three large adoption agencies in Minnesota; non-adoptive families were recruited from Minnesota state birth records to have sibling pairs comparable in age and gender composition to adoptive sibling pairs. Eligibility for adoptive families included having an adopted adolescent placed before the age of 2 years ($M = 4.7$ months, $SD = 3.4$) and between the ages of 11 and 21 at the intake assessment. Adopted and non-adopted siblings were no more than 5 years apart in age (adopted siblings: $M = 2.4$, $SD = 1.0$; non-adopted siblings: $M = 2.1$, $SD = 0.7$) and had no physical, intellectual, or behavioral disabilities that would preclude completing the in-person assessments.

Typical of the Minnesota demographic, much of the sample is of European ancestry (53%); however, many of the adoptees are of Korean descent (70% of adoptees, or 39% of full sample). The remaining 8% of the sample was either African American, Hispanic, or Mixed/Other. Because of the small sample size of these groups, they are coded as “Other.” Fifty-five percent of the sample was female. The age of the offspring when they completed each questionnaire is provided in Table 1.

Table 1

Descriptive statistics for adopted and non-adopted offspring

Variable	Full Sample		Adopted		Non-adopted		Adopted versus Non-adopted	
	<i>M (SD)</i>	<i>N</i>	<i>M (SD)</i>	<i>N</i>	<i>M (SD)</i>	<i>N</i>	<i>p</i>	<i>d</i>
Age – Relationship Quality	17.1 (2.6)		17.3 (2.6)		16.9 (2.5)		.012	0.15
Age – Sexual Behavior	20.6 (2.7)		20.7 (2.6)		20.4 (2.7)		.015	0.15
Sexual Behavior	27.6 (13.0)	1063	28.7 (13.0)	602	26.0 (12.7)	461	<.001	0.21
Parent Report Relationship Quality	124.3 (13.6)	1113	122.2 (13.6)	644	127.05 (13.1)	489	<.001	-0.36
Offspring Report Relationship Quality	121.8 (18.0)	1107	120.1 (18.6)	628	123.95 (17.1)	479	<.001	-0.21

Note. Age – Relationship quality refers to the age of offspring when the parent-offspring relationship quality was reported on. For sexual behavior and parent and offspring reports of parent-offspring relationship quality, values reflect raw sum scores.

Measures

Sexual Behavior

The sexual behavior inventory included 12 items to measure offspring's sexual behavior in casual and committed relationships. This measure was included at Time 2 for adolescents who were at least 16 years old ($n = 433$) at the time of data collection, while the rest were given the measure at Time 3 (time points were approximately 3.5 years apart; $n = 630$), when all participants were at least 19 years old. Across time points, there were 1063 valid reports from offspring in 585 families. Four binary question items (1 = *no*, 6 = *yes*) assessed whether participants had ever engaged in (1) casual oral sex, (2) oral sex while in a committed relationship, (3) casual vaginal/anal sex, and (4) vaginal/anal sex while in a committed relationship. An additional eight items assessed for the same four categories (casual oral, casual vaginal/anal sex, committed oral, committed vaginal/anal sex) (a) the number of partners in the last 12 months, and (b) the number of lifetime partners (e.g., In the casual vaginal/anal sex category of number of lifetime partners: “How many different casual partners have you had sexual intercourse with in your lifetime?”). All eight questions about sexual partner numbers used a 6-point response scale (1 = *None*, 2 = *1-2*, 3 = *3-5*, 4 = *6-9*, 5 = *10-19*, 6 = *20+*). A unidimensionality analysis using the unidim procedure suggested that the 12 items were adequately homogenous and could be treated as a single factor ($u = .85$), and omega hierarchical suggested that the general factor was reliably measured ($\omega_h = .73$, $\omega_t = .96$, $\alpha = .93$)². Given

² Unidim is a new measure of scale homogeneity and unidimensionality [21]. Although Revelle and Condon did not provide cutoff values [21], unidim values for the big five personality traits ranged from .89 to .96 and was .90 for the International Cognitive Ability Resource test. Meanwhile, simulations of orthogonal factors produce unidim values ranging from .12 to .32. Research on the developmental antecedents of sexual behavior also suggests a common “sexual propensity” factor that is accounted for by shared genetic and environmental influences [14,22]. Despite this, because our 12-item sum score operationalization has not been used previously, we conducted several sensitivity analyses in which measures with specific subsets of sexual behavior items were substituted in analyses (e.g., separate casual vs. committed partner item measures, and a measure only using partners in the past year items).

adequate scale homogeneity and reliability of a general factor, we used a sum score of all 12 items as a measure of sexual behavior. Descriptive statistics across and by offspring adoption status are reported in Table 1.

Parent-Offspring Relationship Quality

The Parent Environment Questionnaire (PEQ) was used to measure parent-offspring relationship quality at Time 1 and Time 2 [23]. For offspring who completed the sexual behavior measure at Time 2, their PEQ score at Time 1 was used in analyses, whereas, if offspring completed the sexual behavior measure at Time 3, their Time 2 PEQ score was used. The final measure contains four correlated subscales: conflict, involvement, parent regard for offspring, and offspring regard for parent. The measure was administered to both parents and offspring. At each time, parents reported their relationship with both offspring, and offspring reported their relationship with both parents. The PEQ has been shown to negatively predict externalizing psychopathology [24,25] and shows convergent validity with other established measures of the familial environment [23].

The conflict subscale consists of 12 items that measure disagreement and tension (e.g., “*My parent often criticizes me*”). The involvement subscale has 12 items that measure parent-offspring communication, closeness, and support (e.g., “*I talk about my concerns and my experiences with my parent*”). The parent regard for offspring subscale consists of five items that assess whether the parent is perceived to have regard for the offspring (e.g., “*I know my parent loves me*”). The offspring regard for parent subscale measures the extent to which the offspring respects and is proud of their parent (e.g., “*I respect my parent*”). These four subscales (with conflict reversed) show high positive inter-correlations (parents $r_s = .48-.68$; offspring $r_s = .58-.75$). All items used a 4-point response scale from 1 = *definitely false* to 4 = *definitely true*.

Similar to previous studies, we use a total sum score of the PEQ subscales as a broad measure of parent-offspring relationship quality [23,26].

For offspring reports, the mother and father reports were highly correlated ($r = .80$) and were averaged to form an offspring reported parent-offspring relationship quality measure. Reports from mothers and from fathers were moderately correlated with each other ($r = .31$), and were similarly averaged to form a measure of parent reported parent-offspring relationship quality. Because mother and father reported parent-offspring relationship quality had a substantial amount of unshared variance, we also used their reports separately in additional models. When only one parent report was available—25% of parent reports and 1% of offspring reports, most commonly only a mother's report—the average rating was based on that single rating. Reliability of the composite of the four subscales was good for both parent reports ($u = .98$, $\omega_t = .88$, $\alpha = .83$) and offspring reports ($u = .99$, $\omega_t = .92$, $\alpha = .89$). Although parent reports were moderately correlated with offspring reports ($r = .40$), because parents and offspring are distinct sources of information about the parent-offspring relationship, we used them as predictors in separate models to see if results were consistent across informants.

Statistical Analysis

We regressed out the effects of age and sex on each measure and used the resulting residuals in all statistical analyses. Because the sexual behavior score was markedly positively skewed, it was log transformed prior to age-sex adjustment.

The first step in our analysis examined evidence for a common rearing environmental effect on sexual behavior through biometric analysis of adopted and non-adopted sibling similarity. This analysis estimated sibling correlations, which were then fitted to alternative biometric models. The standard univariate biometric model assumes that the variance in an

observed phenotype can be apportioned among three factors: additive genetic (a^2 or A), shared environmental (c^2 or C , effects shared by reared-together relatives), and non-shared environmental (e^2 or E , effects that differ among reared-together relatives) components. Under standard assumptions, the expected adopted sibling correlation equals c^2 and so provides a direct estimate of the variance attributable to common rearing. Alternatively, the expected non-adopted sibling correlation equals $\frac{1}{2} a^2 + c^2$, and so greater non-adopted than adopted sibling correlation implicates the existence of additive genetic effects. We fit the following alternative biometric models (designated by included components): the full ACE model as well as the AE, CE, and E sub models. (Because the E component also includes effects attributable to measurement error this component must be included in all models.) Relative model fit was evaluated by AIC, with the model having the lowest AIC judged the best-fitting model.

The second step in our analysis involved determining whether the association between the quality of the parent-offspring relationship and sexual behavior reflected, at least in part, environmental processes or alternatively could be attributable entirely to genetic confounding from passive rGE . In non-adopted families, where the genetic parents are also the rearing parents, any association between parent and offspring behavior could owe to shared genetics as well as the shared environment; in contrast, adopted parent-offspring pairs share no segregating genetic factors, which allows us to resolve this form of rGE . If genetic confounding accounts for the correlation between the parent-offspring relationship and sexual behavior, we would expect (1) greater correlation in non-adopted than adopted parent-offspring pairs (reflecting the shared genetic effect), and (2) no correlation in the adopted parent-offspring pairs (reflecting the absence of evidence for shared environmental effects.) We investigated these possibilities both by computing the correlation between parent-offspring relationship quality and sexual behavior

separately in adopted and non-adopted families, and by fitting the mixed-effect models described below.

The final step in our analysis involved investigating whether the parent-offspring relationship is associated with offspring sexual behavior within-families as well as between-families. If the parent-offspring relationship influences sexual behavior then we expect this relationship to hold within-families (i.e., sibling differences in the parent-offspring relationship are associated with sibling differences in sexual behavior) and especially within adopted families (where genetic confounding is absent.) The mixed model analysis involved regressing each sibling's sexual behavior score on the sibling pair mean relationship quality score (between-family effects) and the sibling difference in relationship quality scores (within-family effects). The within- and between-family effects were interacted with adoption status and age, sex, and ethnicity were included as covariates. All analyses were conducted separately using parent-reported and offspring reported measures of parent-offspring relationship quality. Supplemental analyses were conducted to explore whether mother-offspring and father-offspring relationships had similar effects, and whether effects were robust to alternative operationalizations of sexual behavior items.

Results

The *OpenMx R* package was used to fit the biometric sibling model [27]. The *lme4* and *lmerTest* packages were used to fit mixed-effects models [28,29]. We provide additional analysis information on the [Open Science Framework](#) and the data can be accessed upon application request to the MCTFR.

The shared environment

With one exception, all phenotypes showed significant cross-sibling correlations (Table 2). For the sexual behavior phenotype, non-adopted sibling pairs showed a larger correlation than

adopted sibling pairs, consistent with genetic influences on sexual behavior in addition to any environmental effects. For parent-offspring relationship quality, siblings were more similar when the relationship was reported by parents than when it was reported by offspring. However, within each reporter, adopted and non-adopted sibling correlations were similar in magnitude, suggesting that resemblance in relationship quality mainly reflects shared environmental influences rather than genetic factors. These patterns were similar across male-male, female-female, and male-female sibling pairs; correlations stratified by sex and sibling-sex composition are reported in the supplemental materials.

Table 2

Sibling correlations by relationship type

Variable	Adopted			Non-adopted		
	<i>r</i> [95% CI]	<i>df</i>	<i>p</i>	<i>r</i> [95% CI]	<i>df</i>	<i>p</i>
Sexual Behavior	.16 [.05, .26]	315	.005	.31 [.16, .44]	159	<.001
Parent Report Relationship Quality	.31 [.21, .40]	354	<.001	.35 [.22, .47]	182	<.001
Offspring Report Relationship Quality	.13 [.03, .23]	341	.002	.14 [-.01, .28]	175	.061

Note. Adopted siblings included sibling pairs of two adopted offspring or one adopted and one non-adopted offspring. Non-adopted offspring were pairs of non-adopted siblings. All *p* < .05 are bolded.

A single phenotype biometric ACE model on the sexual behavior phenotype fit best and revealed that shared environmental influences accounted for 15%, 95% CI [4%, 25%] of the variance in offspring sexual behavior (Table 3).

Table 3

Test of full and nested biometrical models accounting for variance in offspring sexual behavior

Model	Proportion of variance estimate (95% CI)			Model fit statistics			
	a^2	c^2	e^2	AIC	<i>df</i>	$\Delta - 2LL$	<i>p</i>
ACE	0.32 (0.00, 0.65)	0.15 (0.04, 0.25)	0.53 (0.26, 0.84)	1405	1059		
AE	0.62 (0.33, 0.87)		0.38 (0.13, 0.67)	1410	1060	7.43	.006
CE		0.20 (0.12, 0.29)	0.80 (0.71, 0.88)	1406	1060	3.18	.075
E			1.00	1424	1061	23.36	<.001

Note. The best fitting model by the AIC is bolded.

AIC = Akaike's Information Criterion; $\Delta - 2LL$ = Change in Log Likelihood compared to full ACE model.

Passive gene-environment correlation

For non-adoptees, better-quality parent–offspring relationships were associated with lower sexual behavior (parent-report relationship quality: $r = -.15$, 95% $CI [-.24, -.05]$, $p = .002$; offspring-report relationship quality: $r = -.20$, 95% $CI [-.29, -.11]$, $p < .001$). For adoptees, the pattern was similar (parent-report: $r = -.23$, 95% $CI [-.30, -.15]$, $p < .001$; offspring-report: $r = -.25$, 95% $CI [-.33, -.18]$, $p < .001$), and the correlations did not differ significantly between groups (Fisher’s r -to- z tests, p ’s $> .15$), indicating passive rGE is unlikely to account for the association.

Within-family effects

There was both a significant between-family and within-family effect that was consistent across informants (Table 4). The former effect indicates that families in which parents, on average, have better relationships with their offspring, have offspring with relatively low sexual behavior. The within-family effect suggests that siblings who have a higher quality relationship with their parents have lower sexual behavior relative to their sibling. Further, adoption status did not significantly moderate either the between-family associations ($ps > .16$), supporting the findings from the bivariate correlations that passive rGE does not explain these associations, or the within-family associations ($ps > .56$) consistent with a casual effect within families. In additional analyses, results were unchanged across different operationalizations of sexual behavior, no association was moderated by offspring sex, and associations were highly similar when using mother and father reports as separate predictors. Details of those analyses are presented in supplemental materials.

Table 4

Within- and between-family associations from mixed-effects models regressing offspring sexual behavior on parent reported and offspring reported parent-offspring relationship quality

Association	Model					
	Parent Reports Relationship Quality			Offspring Reports Relationship Quality		
	β	[95% CI]	p	β	[95% CI]	p
Between-Family Relationship Quality						
Main Effect	-0.14	[-0.21, -0.08]	<.001	-0.20	[-0.27, -0.13]	<.001
Interaction with Adoption Status	-0.09	[-0.22, 0.04]	.166	-0.03	[-0.16, 0.10]	.663
Adoptees Simple Effect	-0.19	[-0.28, -0.10]	<.001	-0.21	[-0.30, -0.13]	<.001
Non-adoptees Simple Effect	-0.10	[-0.20, 0.00]	.061	-0.19	[-0.29, -0.08]	.001
Within-Family Relationship Quality						
Main Effect	-0.12	[-0.18, -0.06]	<.001	-0.12	[-0.17, -0.06]	<.001
Interaction with Adoption Status	0.04	[-0.08, 0.15]	.566	-0.02	[-0.13, 0.09]	.748
Adoptees Simple Effect	-0.10	[-0.17, -0.03]	.006	-0.13	[-0.20, -0.05]	.001
Non-adoptees Simple Effect	-0.14	[-0.23, -0.04]	.004	-0.11	[-0.19, -0.02]	.015

Note. Offspring adoption status was effects coded (0.5 = adopted, -0.5 = non-adopted). All associations presented are fully standardized. *ps* <.05 are bolded. Simple effects represent estimated slopes of relationship quality on sexual behavior for adopted and non-adopted offspring.

Discussion

Our findings provide consistent evidence supporting an environmentally driven association between parent-offspring relationship quality and offspring sexual behavior. First, we found a moderate shared environmental influence on offspring sexual behavior, notable in comparison to the minimal shared environmental influences found with most behavioral traits [30]. Our analysis thus suggests that environmental factors (broadly construed) shared by siblings influence their sexual behavior. Second, bivariate correlations indicated that better parent-offspring relationships were similarly associated with lower sexual behavior in both adopted and non-adopted offspring. This suggests that the association between parent-offspring relationships and offspring sexual behavior is not confounded by passive *r*GE, thus,

strengthening the claim that the association is environmentally mediated. Third, parent-offspring relationship quality prospectively predicted lower offspring sexual behavior both between families and within families. The between-family effect that was not moderated by adoption status further supports that relationship quality is an environmental effect that contributes to the shared environmental effect for offspring sexual behavior. The within-family effect, which is not confounded by shared environmental influences further strengthens the claim that parent-offspring relationship quality uniquely influences offspring sexual behavior. Importantly, associations were consistent across both parent and offspring reports of parent-offspring relationship quality. Further, because the data were prospective, reverse causality (reactive *rGE*) is not likely to account for the associations. In sum, our results strengthen the claim that higher quality parent-offspring relationships reduce risky sexual behavior in adolescence.³

Consistent evidence for environmental effects is rare in research on sexual development, as behavioral genetic studies frequently reveal that associations assumed to be causal are confounded by shared genetic and environmental influences. [14,34,35]. A recent review of longitudinal studies on adolescent sexual behavior identified family climate as a particularly promising predictor, whereas evidence for caregiver warmth as a predictor was mixed [36]. Parent-offspring relationship quality does not neatly fit into either category, but the current study suggests that it has “promising evidence” as a predictor of adolescent sexual behavior (the highest category of evidence per Forrester and colleagues) [36].

³ Our measures focus on the number of sexual partners; although it may be contended that this does not measure risky sexual behavior per se, all things considered, having more sexual partners is associated with outcomes such as increased chance of unplanned pregnancy, and contracting a sexually transmitted infection [31]. Nonetheless, having more sexual partners may not be inherently risky, and we encourage researchers to explore these questions with the perspective that many aspects of adolescent and young adult sexual behavior are normative and healthy [32,33].

Limitations and Future Directions

Two major adoption study assumptions (no selective placement or range restriction) are unlikely to have had a large impact on our findings. First, the adoptees were predominantly international placements with little information on birth background limiting the possibility for selective placement [20,37]. Second, range restriction would be evident in lower environmental variability in adopted as compared to non-adopted families, but we found the reverse. Thus, our findings strongly suggest that passive rGE is unlikely to explain our results. In addition, our prospective design reduced the chance that prior offspring sexual behavior was influencing parent-offspring relationships. Nonetheless, we cannot rule out reverse causality because sexual behavior prior to assessment of parent-offspring relationship quality could account for the associations. Future research should provide more stringent tests of reverse causality via longitudinal models that eliminate these alternative hypotheses while maintaining control for genetic confounding (e.g., a monozygotic twin difference cross-lagged panel model) [38].

Additionally, while our study suggests the existence of an environmental effect, we cannot adjudicate between different hypotheses for *how* this environmental effect arises. One pathway through which better parent-offspring relationships may exert long-term effects on sexual behavior is by promoting more stable and enduring romantic relationships into young adulthood [39,40]. Sibling dynamics may also play a role; for example, younger siblings may model the sexual behavior of older siblings who serve as salient social models. Future research that can adjudicate between different process-based hypotheses will be important to inform potential interventions.

Conflict of Interest Disclosure

All authors declare that they have no real or perceived conflicts of interest related to the conduct or publication of this research.

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Use of AI

During the preparation of this work, the first author used *ChatGPT (OpenAI)* to assist with grammar and word choice. After using this tool, the first author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

- [1] Deptula DP, Henry DB, Schoeny ME. How can parents make a difference? Longitudinal associations with adolescent sexual behavior. *J Fam Psychol.* 2010;24(6):731-739. doi:10.1037/a0021760
- [2] Johnson MD. Parent–Child Relationship Quality Directly and Indirectly Influences Hooking Up Behavior Reported in Young Adulthood through Alcohol Use in Adolescence. *Arch Sex Behav.* 2013;42(8):1463-1472. doi:10.1007/s10508-013-0098-9
- [3] Kan ML, Cheng Y hsin A, Landale NS, McHale SM. Longitudinal Predictors of Change in Number of Sexual Partners across Adolescence and Early Adulthood. *J Adolesc Health.* 2010;46(1):25-31. doi:10.1016/j.jadohealth.2009.05.002
- [4] Rink E, Tricker R, Harvey SM. Onset of Sexual Intercourse among Female Adolescents: The Influence of Perceptions, Depression, and Ecological Factors. *J Adolesc Health.* 2007;41(4):398-406. doi:10.1016/j.jadohealth.2007.04.017
- [5] Roche KM, Ahmed S, Blum RW. Enduring consequences of parenting for risk behaviors from adolescence into early adulthood. *Soc Sci Med.* 2008;66(9):2023-2034. doi:10.1016/j.socscimed.2008.01.009
- [6] Vasilenko SA. Sexual Behavior and Health From Adolescence to Adulthood: Illustrative Examples of 25 Years of Research From Add Health. *J Adolesc Health.* 2022;71(6):S24-S31. doi:10.1016/j.jadohealth.2022.08.014
- [7] Fraley RC, Roisman GI, Haltigan JD. The legacy of early experiences in development: Formalizing alternative models of how early experiences are carried forward over time. *Dev Psychol.* 2013;49(1):109-126. doi:10.1037/a0027852
- [8] Bronfenbrenner U. Contexts of Child Rearing: Problems and prospects. *Am Psychol.* 1979;34(10):844-850.
- [9] Maccoby EE, Eleanor E. Maccoby. Lindzey G, ed. 1992;28(6):290-335. doi:10.1037/11347-009
- [10] Scarr S, McCartney K. How people make their own environments: A theory of genotype→ environment effects. *Child Dev.* Published online 1983:424-435. Accessed June 10, 2024. <https://psycnet.apa.org/record/1983-25048-001>
- [11] Plomin R, DeFries JC, Loehlin JC. Genotype-environment interaction and correlation in the analysis of human behavior. *Psychol Bull.* 1977;84(2):309. Accessed June 7, 2024. <https://psycnet.apa.org/record/1977-24913-001>
- [12] Kendler KS, Baker JH. Genetic influences on measures of the environment: a systematic review. *Psychol Med.* 2007;37(05):615. doi:10.1017/S0033291706009524

- [13] Barbaro N, Boutwell BB, Barnes JC, Shackelford TK. Genetic confounding of the relationship between father absence and age at menarche. *Evol Hum Behav.* 2017;38(3):357-365. doi:10.1016/j.evolhumbehav.2016.11.007
- [14] Samek DR, Iacono WG, Keyes MA, Epstein M, Bornovalova MA, McGue M. The developmental progression of age 14 behavioral disinhibition, early age of sexual initiation, and subsequent sexual risk-taking behavior. *J Child Psychol Psychiatry.* 2014;55(7):784-792. doi:10.1111/jcpp.12176
- [15] Neiderhiser JM, Reiss D, Pedersen NL, et al. Genetic and environmental influences on mothering of adolescents: a comparison of two samples. *Dev Psychol.* 2004;40(3):335. Accessed June 7, 2024. <https://psycnet.apa.org/record/2004-13591-002>
- [16] Munafò MR, Higgins JPT, Smith GD. Triangulating Evidence through the Inclusion of Genetically Informed Designs. *Cold Spring Harb Perspect Med.* 2021;11(8):a040659. doi:10.1101/cshperspect.a040659
- [17] Lahey BB, D'Onofrio BM. All in the Family: Comparing Siblings to Test Causal Hypotheses Regarding Environmental Influences on Behavior. *Curr Dir Psychol Sci.* 2010;19(5):319-323. doi:10.1177/0963721410383977
- [18] McGue M, Osler M, Christensen K. Causal Inference and Observational Research: The Utility of Twins. *Perspect Psychol Sci.* 2010;5(5):546-556. doi:10.1177/1745691610383511
- [19] Revelle W. psych: Procedures for Psychological, Psychometric, and Personality Research. Published online 2025. <https://cran.r-project.org/package=psych>
- [20] McGue M, Keyes M, Sharma A, et al. The Environments of Adopted and Non-adopted Youth: Evidence on Range Restriction From the Sibling Interaction and Behavior Study (SIBS). *Behav Genet.* 2007;37(3):449-462. doi:10.1007/s10519-007-9142-7
- [21] Revelle W, Condon D. Unidim: An index of scale homogeneity and unidimensionality. *Psychol Methods.* Published online March 3, 2025. doi:10.1037/met0000729
- [22] Clark DA, Durbin CE, Heitzeg MM, Iacono WG, McGue M, Hicks BM. Sexual Development in Adolescence: An Examination of Genetic and Environmental Influences. *J Res Adolesc.* 2020;30(2):502-520. doi:10.1111/jora.12540
- [23] Elkins IJ, McGue M, Iacono WG. Genetic and environmental influences on parent-son relationships: Evidence for increasing genetic influence during adolescence. *Dev Psychol.* 1997;33(2):351-363. doi:10.1037/0012-1649.33.2.351
- [24] Burt SA, Krueger RF, McGue M, Iacono W. Parent-Child Conflict and the Comorbidity Among Childhood Externalizing Disorders. *Arch Gen Psychiatry.* 2003;60(5):505. doi:10.1001/archpsyc.60.5.505

- [25] Klahr AM, McGue M, Iacono WG, Burt SA. The association between parent–child conflict and adolescent conduct problems over time: Results from a longitudinal adoption study. *J Abnorm Psychol.* 2011;120(1):46-56. doi:10.1037/a0021350
- [26] Ludeke S, Johnson W, McGue M, Iacono WG. Genetic amplification and the individualization of the parent–child relationship across adolescence. *Psychol Med.* 2013;43(2):413-422. doi:10.1017/S0033291712001201
- [27] Neale MC, Hunter MD, Pritikin JN, et al. OpenMx 2.0: Extended structural equation and statistical modeling. *Psychometrika.* 2016;81:535-549. Accessed June 5, 2024. https://idp.springer.com/authorize/casa?redirect_uri=https://link.springer.com/article/10.1007/s11336-014-9435-8&casa_token=L55Jf51s7NwAAAAA:hwA4sqUnnXISuagYbxNcqI_eIsp-72a1Sq7KhHM-sZiCyDyRZ73vQfIBdPSKc0MJr5-EHs2-czdazFk
- [28] Bates D, Mächler M, Bolker B, Walker S. Fitting Linear Mixed-Effects Models Using lme4. *J Stat Softw.* 2015;67(1). doi:10.18637/jss.v067.i01
- [29] Kuznetsova A, Brockhoff PB, Christensen RHB. **lmerTest** Package: Tests in Linear Mixed Effects Models. *J Stat Softw.* 2017;82(13). doi:10.18637/jss.v082.i13
- [30] Bouchard TJ, McGue M. Genetic and environmental influences on human psychological differences. *J Neurobiol.* 2003;54(1):4-45. doi:10.1002/neu.10160
- [31] Kelley S, Borawski E, Flocke S, Keen K. The role of sequential and concurrent sexual relationships in the risk of sexually transmitted diseases among adolescents. *J Adolesc Health.* 2003;32(4):296-305. doi:10.1016/S1054-139X(02)00710-3
- [32] Harden KP. A Sex-Positive Framework for Research on Adolescent Sexuality. *Perspect Psychol Sci.* 2014;9(5):455-469. doi:10.1177/1745691614535934
- [33] Tolman DL, McClelland SI. Normative Sexuality Development in Adolescence: A Decade in Review, 2000–2009. *J Res Adolesc.* 2011;21(1):242-255. doi:10.1111/j.1532-7795.2010.00726.x
- [34] Huibregtse BM, Bornovalova MA, Hicks BM, McGue M, Iacono W. Testing the Role of Adolescent Sexual Initiation in Later-Life Sexual Risk Behavior: A Longitudinal Twin Design. *Psychol Sci.* 2011;22(7):924-933. doi:10.1177/0956797611410982
- [35] Huibregtse BM, Hatoum AS, Corley RP, Rhea SA, Hewitt JK, Stallings MC. Etiological Overlap Between Sex Under the Influence and Number of Lifetime Sexual Partners. *Behav Genet.* 2021;51(1):12-29. doi:10.1007/s10519-020-10019-7
- [36] Forrester EJ, Henke J, Knab J, Cole R. Promising Predictors of Adolescent Sexual Behavior Outcomes: A Systematic Review of Longitudinal Research. *J Adolesc Health.* Published online January 2025:S1054139X24008243. doi:10.1016/j.jadohealth.2024.12.002
- [37] Willoughby EA, Edwards T, Lee JJ. Durable lessons on intelligence from twin and adoption research. *Manuscr Rev Behav Genet.* Published online 2025.

- [38] Kullberg MJ, Van Schie CC, Allegrini AG, et al. Comparing findings from the random-intercept cross-lagged panel model and the monozygotic twin difference cross-lagged panel model: Maladaptive parenting and offspring emotional and behavioural problems. *JCPP Adv.* 2024;4(1):e12203. doi:10.1002/jcv2.12203
- [39] Lamb PS, Dagan O, Dugan KA, Bleil ME, Booth-LaForce C, Roisman GI. Childhood interpersonal antecedents of adult romantic relationship adjustment: Prospective evidence from the National Institute of Child Health and Human Development Study of Early Child Care and Youth Development. *J Fam Psychol.* Published online September 11, 2025. doi:10.1037/fam0001398
- [40] Roisman GI, Fraley RC. A Behavior-Genetic Study of the Legacy of Early Caregiving Experiences: Academic Skills, Social Competence, and Externalizing Behavior in Kindergarten. *Child Dev.* 2012;83(2):728-742. doi:10.1111/j.1467-8624.2011.01709.x